

What does "Prophet" mean?

The word **Prophet** means:

A person who predicts or tells what may happen in the future.

Examples:

- Weather forecaster predicts rain.
- Economist predicts market growth.
- Prophet predicts future events.

In Time Series:

Prophet predicts future values using historical data.

That's why Meta named the forecasting model "**Prophet**".

Who Invented Prophet?

Prophet was developed by:

Meta Platforms (formerly Facebook)

Around **2017**, Facebook's Core Data Science Team officially released Prophet as an open-source forecasting tool.

Why Did Facebook Create Prophet?

The Problem

Facebook had thousands of time series datasets:

- Daily active users
- Website traffic
- Advertisement clicks
- User engagement

- Revenue data

Data scientists faced challenges:

- ✗ ARIMA required stationarity
 - ✗ Parameter tuning was difficult
 - ✗ Business analysts couldn't easily use forecasting models
 - ✗ Handling holidays and seasonal events was complex
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Facebook's Goal

They wanted a model that:

- ✓ Works automatically
- ✓ Handles seasonality
- ✓ Handles holidays
- ✓ Requires minimal tuning
- ✓ Can be used by non-experts

Thus, **Prophet** was born.

Real Story Behind Prophet

Before Prophet:

Business Team



Data Scientist



ARIMA Model



Forecast

This process was slow.

Facebook wanted:

Business Team



Prophet



Forecast

Much faster and easier.

Why the Name "Prophet"?

Think of a fortune teller:

Past Events



Understand Patterns



Predict Future

Prophet does the same with data.

Example:

Past Website Visitors



Learns Trend



Predicts Future Visitors

Hence the name:

"Prophet"

= Predictor of future events.

First Public Release

Prophet was publicly released in **2017** as an open-source project.

It supports:

- Python
- R

and became popular because it was much easier than traditional forecasting methods.

How Prophet Works

Prophet assumes:

Forecast

=

Trend

+

Seasonality

+

Holiday Effects

+

Error

Example:

Website Traffic

Traffic Growth

+

Weekend Effect

+

Festival Traffic Spike

+

Random Noise

Why Companies Use Prophet

E-Commerce

Predict sales.

Hotels

Predict bookings.

Airlines

Predict passengers.

Banks

Predict transactions.

Websites

Predict visitor traffic.

Retail Stores

Predict demand.

Interview Question

Mam: Why is it called Prophet?

Answer:

Prophet is a forecasting model developed by Facebook (Meta) in 2017. It was named Prophet because, like a prophet who predicts future events, the model predicts future values by learning patterns from historical data. It automatically captures trends, seasonality, and holiday effects, making forecasting simple and accurate.

Easy Memory Trick

Remember:

ARIMA
=
Statistician

Prophet
=
Fortune Teller

Both predict the future, but Prophet was designed to make forecasting easier and more automated for business users.

This story is often appreciated in project reviews because it shows you understand **why the algorithm was created**, not just how to run the code.

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Benefits of Website Traffic Forecasting Using Prophet

If your mam asks "**Why did you do this project? What is the business benefit?**", you can answer like this:

1. Predict Future Visitors

The main goal is to estimate how many people will visit a website in the future.

Example:

Date	Predicted Visitors
Tomorrow	2,500
Next Week	18,000
Next Month	75,000

This helps businesses prepare in advance.

2. Digital Marketing Planning

If traffic is expected to decrease:

- Run advertisements
- Send email campaigns
- Launch offers

If traffic is expected to increase:

- Increase marketing budget
 - Launch new products
-

3. Server Resource Planning

High traffic can slow down websites.

Forecasting helps companies:

- Upgrade servers
- Increase cloud resources
- Avoid website crashes

Example:

Amazon, Flipkart, Myntra prepare servers before festival sales.

4. Revenue Prediction

For websites earning through:

- Ads
- Affiliate marketing
- E-commerce sales

More visitors generally mean more revenue.

Example:

10,000 visitors/day

↓

Expected revenue ₹5,000/day

5. Identify Trends and Seasonality

Prophet automatically detects patterns like:

Traffic increases every Monday
Traffic decreases every Sunday
Traffic spikes during festivals

This helps management understand user behavior.

6. Better Business Decisions

Forecast results help decide:

- Marketing budget
 - Staffing requirements
 - Product launches
 - Promotional campaigns
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Technical Benefits

Why Prophet?

- ✓ Easy to implement
 - ✓ Handles missing values
 - ✓ Detects seasonality automatically
 - ✓ No need for stationarity testing
 - ✓ Produces understandable visualizations
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Interview Answer (1 Minute)

Website Traffic Forecasting using Prophet helps predict future website visitors based on historical traffic data. The forecast enables businesses to plan marketing campaigns, optimize server resources, estimate future revenue, and understand seasonal traffic patterns. Prophet is particularly useful because it automatically handles trends and seasonality without requiring complex preprocessing.

